

Satellite Image Processing for Urban Expansion Monitoring

Problem Statement

Urban expansion is quickly turning natural land, such as farmland, forests, and open spaces, into developed areas. This growth puts more pressure on the environment and resources. It causes problems like losing green spaces, shrinking water bodies, and increased pollution. Traditional monitoring relies on manual comparison of images from satellites. This is slow, labour-intensive and error-prone. It does not cope well with large sets of data and is not capable of delineating temporal change. We need an AI-based automated system for processing satellite imaging and automatically identifying urban expansion using image processing techniques. Using models like YOLOv11, the system now generates faster and more uniform results than previously.

Motivation

- Cities tend to rise at a fast pace and normally at the cost of nature and greenery, and thus monitoring these changes is of paramount significance to ensure sustainable progress.
- The comparison of satellite pictures by hand is slow and error-prone. AI image processing makes it faster, automatic, and precise.
- A small project like this gives us a chance to use AI and image processing in practice on an everyday problem that affects urban planning and the environment directly.

Objectives

- To preprocess and prepare satellite images for detecting urban features.
- To apply YOLOv11 object detection for identifying buildings and constructions from images.
- To analyze changes in urban areas by comparing images from different time periods.
- To generate accurate results that help in understanding the pattern of urban expansion.

Background

The cities have been growing rapidly due to increases in population, industrial developments, and expansion of infrastructure. The increase mostly yields environmental problems such as deterioration of green spaces, degradation of water bodies, and rises in the level of pollutants.

Satellite imagery provides a useful way of following such changes over a period because it consists of high-level observations of land use and urban developments. The problem, however, lies in interpreting such imagery manually because of the amount of information and complexity of urban landscapes.

Future advances in Artificial Intelligence, specifically object detection models like YOLOv11, have now made it possible to automatically detect building and construction from satellites. By comparison of photographs from different years, we get to trace how cities have been expanding and how they have been impacting the environment.

This mini-project is regarding the usage of satellite imaging and YOLOv11 image processing for urban expansion detection and generating accurate results for supporting city growth patterns

Literature review

No.	Paper Title	Method / Technology Used	Study Area / Dataset	Main Findings
1	Detecting Urban Changes with Recurrent Neural Networks from Multitemporal Sentinel-2 Data	RNN + U-Net	Sentinel-2 satellite images (Europe)	Achieved >95% accuracy in detecting urban growth using temporal patterns in data.
2	Satellite Image Classification with Deep Learning	CNN (Convolutional Neural Network)	Global facility dataset (satellite images)	Classified 63 types of facilities (buildings, roads, airports, etc.) with ~83% accuracy, showing CNN's strength in image recognition.
3	AI-Driven Change Detection in Satellite Imagery: Enhancing GIS Applications	CNNs + GIS integration	Multi-source satellite data	AI detected urbanisation and deforestation more accurately than traditional techniques; results were integrated into GIS for real-time monitoring.
4	Continuous Urban Change Detection from Satellite Image Time Series	Transformers + Temporal Feature Refinement	Multi-year satellite time-series	Proposed continuous monitoring approach instead of before–after images; improved detection of gradual changes in urban areas.
5	Evidence-Based Inference and Quantification of Urban Expansion Using YOLOv8 and High-Resolution Satellite Imagery	YOLOv8 + Google Earth Pro + GIS	Greater Faridabad, India (2014–2025)	Detected buildings & roads with 98% accuracy; quantified ~450% urban growth over 11 years; provided GIS-ready outputs.
6	Deep Learning-Based Change Detection in Remote Sensing Images: A Review	CNN, RNN, GANs (survey)	Many datasets (SAR, multispectral, hyperspectral)	Summarized strengths & weaknesses of AI models; deep learning significantly outperforms older pixel-based techniques.
7	AI for Environmental Sustainability: Advances, Challenges, and Future Directions	Broad AI (ML, DL, GIS, IoT)	Global perspective	Highlighted AI use in climate modeling, biodiversity, pollution monitoring, renewable energy; discussed ethical and data challenges.
8	Urban Expansion Prediction and Land Use/Land Cover Modeling	Editorial overview, policy analysis, hybrid & AI techniques	Global overview / Special Issue	Highlights integration of AI in urban expansion research; emphasizes need for holistic sustainable frameworks and advances in hybrid modeling techniques

9	Monitoring and Forecasting of Urban Expansion Using Machine Learning-Based Techniques	SVM classification, Logistic Regression-Markov chain, LCM, Fuzzy TOPSIS	Gharbia Governorate, Egypt	Urban area projected to increase from 7.8% to 15.2% by 2018; prediction shows continued growth to 33% by 2048; high accuracy (OA > 93%, Kappa > 0.7)
10	Neural Network-Based Urban Change Detection with Deep-Temporal Multispectral and SAR Remote Sensing	Ensemble Recurrent CNN with multispectral & SAR data, synthetic labels	Rotterdam, Liège, Limassol (Europe), multiple datasets	Automated large-scale continuous urban change detection; improved accuracy and robustness through deep temporal modeling; efficient cross-mission modeling
11	Continuous Urban Change Detection from Satellite Images with Temporal Feature Refinement and Multi-Task Integration	ConvNet + Transformer + Markov network; multi-task learning	SpaceNet 7, WUSU, TSCD datasets (global, China)	Outperforms SOTA on continuous urban change detection; achieves higher F1, IoU, accuracy; preserves temporal dimension for improved change mapping
12	A Novel CNN-LSTM-based Approach to Predict Urban Expansion	CNN-LSTM (ConvLSTM), unsupervised semantic segmentation	Riyadh, Jeddah, Dammam (Saudi Arabia)	Achieves 92% classification accuracy; outperforms GAN methods on RMSE, MSE, SSIM, PSNR metrics; suitable for urban growth

Research Gaps

From the examined works, the following key gaps are revealed:

Complexity of Models – Models utilizing RNNs, Transformers, or both achieve extremely good rates of accuracy but require extremely large datasets and intensive computation that are too large for a mini-project.

GIS Integration Dependence – Other projects rely on integration with GIS systems to support real-time mapping and decision support. Although powerful, this adds implementation complexity not suitable for a lightweight system.

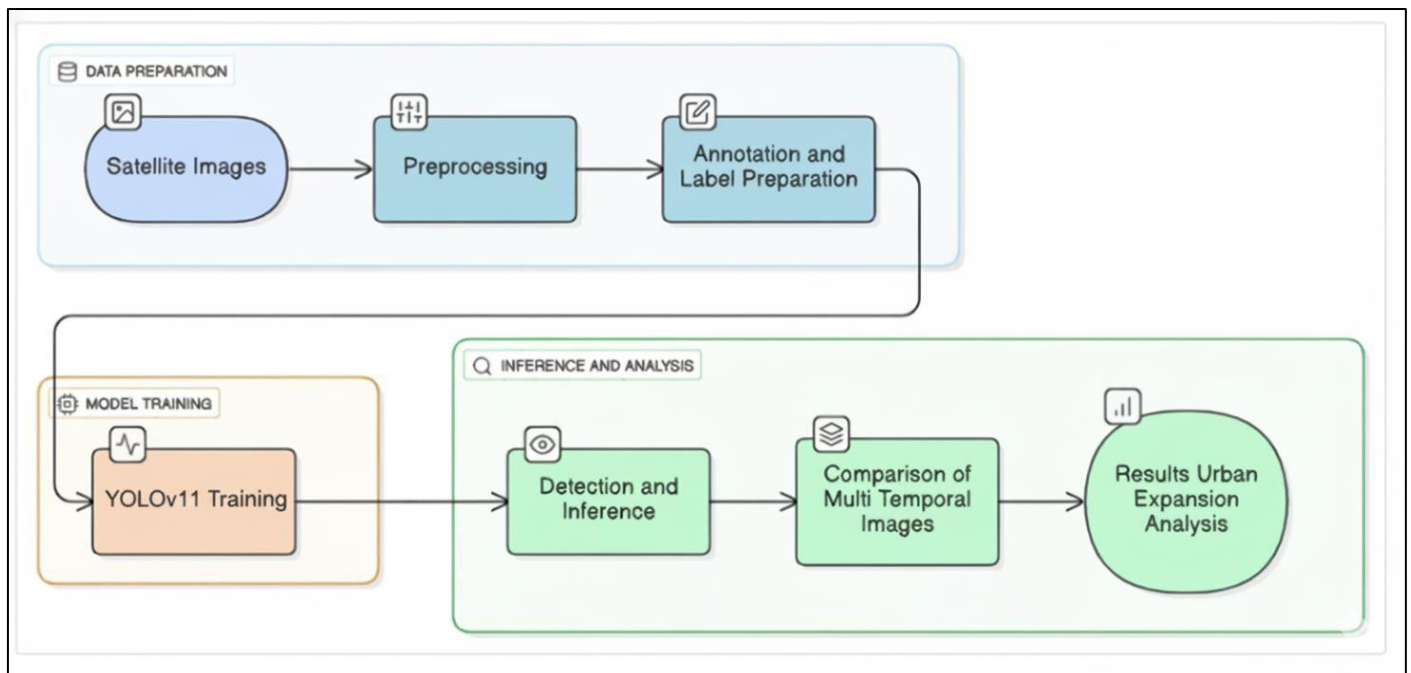
Temporal Data Handling – The camera systems function on multi-year long sequences of images and necessitate long-duration data handling and preprocessing pipelines.

Generalization vs. Practicality – Reviews mention that the deep learning models generalize well on different datasets, whereas simpler, direct-to-output approaches (e.g., object detection for a small dataset) are more translatable to demonstrable mini-projects.

Methodology

Satellite Images → Preprocessing → YOLOv11 Training → Detection → Change Detection → Results & Visualization

The methodology of the proposed system is divided into structured phases to ensure effective detection and analysis of urban expansion using satellite imagery. The overall workflow is shown in the block diagram below :



1. Data Collection

- **Dataset used:** SpaceNet-7: Multi-Temporal Urban Development Dataset — contains satellite images of identical geographic regions captured over multiple years.
- **Data type:** Multi-temporal satellite imagery representing the progression of urbanization.
- **Focus feature:** Extraction of urban structures such as buildings, roads, and dense built-up zones that act as indicators of urban expansion.

2. Preprocessing

- **Image preparation:**
 - Converted GeoTIFF satellite images into standard formats (.jpg/.png).
 - Resized all images to 640×640 pixels to match YOLOv11's input resolution.
- **Automated feature extraction:**

Applied a combination of image filters to enhance urban structure visibility:

 1. **Canny Edge Detection:** Captures boundaries and outlines of structures.
 2. **HSV Color Filtering:** Isolates gray and brown tones typical of built-up areas.
 3. **Morphological Gradient Filtering:** Highlights texture-rich, densely built regions.
- **Annotation generation:**

Bounding boxes were automatically generated around detected regions and converted into YOLO label format (class_id, x_center, y_center, width, height).
- **Data augmentation:**

Employed Albumentations techniques (horizontal flip, brightness/contrast adjustment, Gaussian noise, blur) to enhance model robustness.
- **Dataset split:**

Divided into training (816 images), validation (204), and testing (180) subsets.

3. Model Training

- **Model used:** YOLOv11m (Medium variant) — latest Ultralytics architecture, optimized for precision and small-object recognition.
- **Training input:** Preprocessed, auto-labeled urban imagery.
- **Training configuration:**

- **Epochs:** 40
- **Batch size:** Auto-adjusted based on GPU capacity
- **Task:** Object detection of urban and built-up features
- **Output:** Trained YOLOv11 model (best.pt) capable of detecting urban structures in unseen satellite images.

4. Object Detection and Automated Analysis

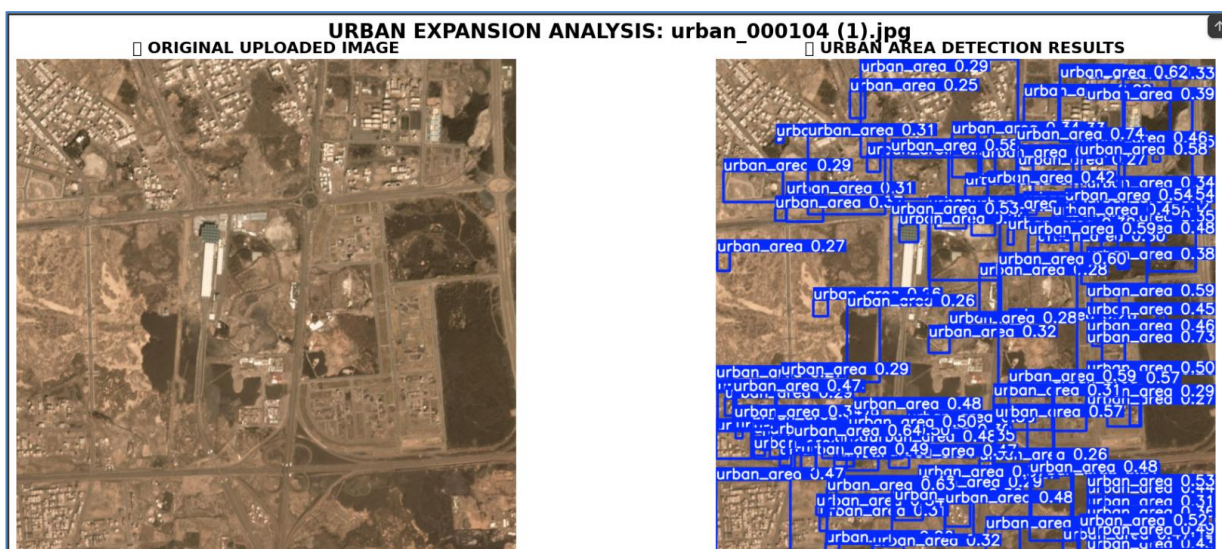
- **Inference workflow:**
A single satellite image is passed through the trained YOLOv11 model, which outputs:
 - Bounding boxes for detected urban features
 - Confidence scores
 - Pixel-based area coverage metrics
- **Automated report generation:**
A custom analytical script (AdvancedUrbanTester) processes the YOLO output to compute and summarize:
 - Total number of detected urban regions
 - Average detection confidence
 - Cumulative urban coverage (as % of total image)
 - Average area size per detection
 - Overall urban expansion classification — categorized as Low, Moderate, High, or Extreme

5. Urban Expansion Interpretation

- Each image is analyzed independently, generating a structured report with quantitative and qualitative indicators of urbanization.
- For multi-temporal study, results from images of different time periods (same region) can be compared manually or statistically to assess temporal urban growth trends.

6. Results and Visualization

- **Outputs generated:**
 - Detected images with bounding boxes highlighting built-up regions.
 - Structured JSON reports containing coverage, confidence, and detection metrics.
- **Interpretation:**
The model effectively identifies and quantifies urban regions, enabling data-driven assessment of urban sprawl, infrastructure density, and environmental implications.



Conclusion

The study rightly demonstrates the application of deep learning techniques through the YOLOv11 model on satellite scenes for monitoring urbanization. Through the recognition and assessment of the urban features such as buildings and built-up areas in multi-temporal scenes, the system provides evident indications of the patterns through which cities evolve over time.

The findings demonstrate the effectiveness of YOLOv11 in dealing with challenging satellite images and producing precise detections through minimal preprocessing. It is swifter than human inspection or conventional classification techniques and scalable and reliable when contrasted with them.

Even if such a mini-project involves only image processing and visualization of the result, the developed framework represents a good starting point for future extensions, e.g., the integration of GIS platforms, multi-class detection (roads, vegetation, water bodies), or real-time monitoring.

Finally, the research demonstrates that the analysis of satellite images using AI constitutes a useful and effective tool to monitor urbanization and help urban planners devise better urbanization plans for the future.

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